

## TRANGIE, Australia

WMO: 957100

Lat: 31.9861S

Lon: 147.9489E

Elev: 216

StdP: 98.76

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>7      | (b)<br>0.5 | (c)<br>1.7 | (d)<br>-6.7                   | (e)<br>2.2 | (f)<br>21.2 | (g)<br>-4.9 | (h)<br>2.6 | (i)<br>19.4 | (j)<br>9.5            | (k)<br>13.0 | (l)<br>8.6 | (m)<br>13.3 | (n)<br>1.4            | (o)<br>150 | (p)<br>0.464 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>1      | (b)<br>13.7            | (c)<br>39.3     | (d)<br>19.7 | (e)<br>37.1 | (f)<br>19.4 | (g)<br>35.4 | (h)<br>19.1 | (i)<br>23.0         | (j)<br>30.0 | (k)<br>22.2 | (l)<br>29.7 | (m)<br>21.5 | (n)<br>29.3 | (o)<br>5.2           | (p)<br>330 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.5                            | 16.5 | 24.5 | 20.3 | 15.4 | 24.0 | 19.3 | 14.5 | 23.9 | 69.2          | 29.4 | 66.2 | 29.6 | 63.4 | 29.4 | 25.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.2              | 8.6  | 7.4 | -2.1                       | 42.3 | 1.0                | 2.2 | -2.8                                               | 43.9 | -3.4       | 45.2 | -3.9       | 46.4 | -4.6       | 48.1 |
|                   |      |     | -2.5                       | 24.2 | 1.0                | 0.7 | -3.2                                               | 24.7 | -3.8       | 25.1 | -4.4       | 25.6 | -5.1       | 26.1 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.4  | 27.2  | 25.9  | 23.0  | 18.9  | 14.0  | 10.9  | 10.0  | 11.1  | 14.5  | 18.6  | 22.2  | 24.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 6.83  | 3.61  | 3.65  | 3.27  | 3.32  | 2.98  | 2.54  | 2.53  | 2.77  | 3.46  | 3.91  | 4.01  | 3.70  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 76    | 0     | 0     | 0     | 0     | 3     | 19    | 32    | 18    | 3     | 0     | 0     | 0     | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1064  | 0     | 1     | 4     | 32    | 138   | 223   | 259   | 224   | 124   | 46    | 10    | 2     | (9)  |
| (10) |                                                                                  | CDD10.0     | 3136  | 533   | 445   | 403   | 268   | 127   | 46    | 31    | 53    | 137   | 267   | 367   | 459   | (10) |
| (11) |                                                                                  | CDD18.3     | 1081  | 274   | 212   | 149   | 50    | 2     | 0     | 0     | 1     | 9     | 54    | 127   | 203   | (11) |
| (12) |                                                                                  | CDH23.3     | 12960 | 3504  | 2478  | 1494  | 441   | 25    | 0     | 0     | 7     | 166   | 711   | 1596  | 2539  | (12) |
| (13) |                                                                                  | CDH26.7     | 6487  | 1993  | 1322  | 644   | 111   | 1     | 0     | 0     | 1     | 42    | 265   | 766   | 1344  | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.5   | 4.1   | 3.9   | 3.7   | 3.1   | 3.0   | 3.0   | 2.9   | 3.2   | 3.5   | 3.8   | 4.1   | 4.2   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 535   | 69    | 46    | 40    | 46    | 49    | 29    | 35    | 38    | 36    | 53    | 46    | 46    | (15) |
| (16) |                                                                                  | PrecMax     | 963   | 218   | 301   | 165   | 318   | 180   | 84    | 98    | 85    | 102   | 250   | 175   | 179   | (16) |
| (17) |                                                                                  | PrecMin     | 227   | 0     | 0     | 1     | 0     | 3     | 2     | 0     | 0     | 0     | 8     | 2     | 0     | (17) |
| (18) |                                                                                  | PrecStd     | 166   | 63    | 58    | 42    | 66    | 45    | 22    | 25    | 24    | 29    | 47    | 40    | 40    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 41.9  | 41.6  | 36.5  | 32.2  | 25.7  | 20.6  | 20.7  | 24.4  | 30.7  | 34.5  | 39.5  | 40.4  | (19) |
| (20) |                                                                                  |             | MCWB  | 20.8  | 21.4  | 19.4  | 17.4  | 15.5  | 13.4  | 11.6  | 13.6  | 14.8  | 16.3  | 18.7  | 18.7  | (20) |
| (21) |                                                                                  | 2%          | DB    | 39.4  | 38.4  | 34.3  | 29.5  | 23.9  | 18.9  | 18.9  | 21.9  | 27.3  | 31.9  | 35.9  | 37.4  | (21) |
| (22) |                                                                                  |             | MCWB  | 20.1  | 20.1  | 18.7  | 17.3  | 15.0  | 13.0  | 11.4  | 12.5  | 14.1  | 16.0  | 17.8  | 18.6  | (22) |
| (23) |                                                                                  | 5%          | DB    | 37.2  | 35.9  | 32.2  | 27.7  | 22.3  | 17.6  | 17.2  | 19.7  | 25.0  | 29.6  | 33.5  | 35.5  | (23) |
| (24) |                                                                                  |             | MCWB  | 20.2  | 19.7  | 18.3  | 16.4  | 14.1  | 12.4  | 11.2  | 11.8  | 13.3  | 15.6  | 17.2  | 18.4  | (24) |
| (25) |                                                                                  | 10%         | DB    | 35.3  | 33.7  | 30.3  | 26.0  | 20.5  | 16.2  | 15.7  | 17.9  | 22.7  | 27.4  | 31.1  | 33.4  | (25) |
| (26) |                                                                                  |             | MCWB  | 19.8  | 19.5  | 18.2  | 15.9  | 13.3  | 11.7  | 10.8  | 11.2  | 13.0  | 15.4  | 17.1  | 18.0  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 23.8  | 24.1  | 22.8  | 20.2  | 17.6  | 15.9  | 15.2  | 16.3  | 17.8  | 20.4  | 21.7  | 22.7  | (27) |
| (28) |                                                                                  |             | MADB  | 30.7  | 30.6  | 28.5  | 25.2  | 21.8  | 18.2  | 16.9  | 19.4  | 24.3  | 25.2  | 29.2  | 28.1  | (28) |
| (29) |                                                                                  | 2%          | WB    | 22.9  | 23.1  | 21.3  | 18.9  | 16.2  | 14.5  | 13.6  | 14.4  | 16.2  | 18.7  | 20.6  | 21.7  | (29) |
| (30) |                                                                                  |             | MADB  | 31.1  | 29.5  | 27.4  | 24.9  | 21.0  | 16.6  | 15.9  | 19.0  | 22.2  | 25.4  | 28.3  | 28.6  | (30) |
| (31) |                                                                                  | 5%          | WB    | 22.2  | 22.2  | 20.4  | 17.9  | 15.1  | 13.5  | 12.5  | 13.0  | 15.1  | 17.8  | 19.7  | 20.9  | (31) |
| (32) |                                                                                  |             | MADB  | 30.9  | 29.1  | 27.6  | 24.3  | 20.5  | 16.1  | 15.5  | 18.1  | 21.1  | 24.8  | 27.1  | 28.3  | (32) |
| (33) |                                                                                  | 10%         | WB    | 21.5  | 21.3  | 19.6  | 17.0  | 14.2  | 12.5  | 11.4  | 11.8  | 14.1  | 16.8  | 18.7  | 20.1  | (33) |
| (34) |                                                                                  |             | MADB  | 30.4  | 28.9  | 27.2  | 23.5  | 19.3  | 15.4  | 14.8  | 16.8  | 20.7  | 24.4  | 26.5  | 27.6  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               |             |       | MDBR  | 13.7  | 12.8  | 12.6  | 12.9  | 12.5  | 10.5  | 11.7  | 13.1  | 14.5  | 14.4  | 14.1  | (35) |
| (36) |                                                                                  | 5% DB       | MADB  | 15.8  | 15.6  | 15.0  | 14.2  | 14.0  | 12.1  | 13.5  | 15.4  | 17.7  | 17.6  | 17.3  | 16.4  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.6   | 4.8   | 5.5   | 5.4   | 6.6   | 6.8   | 7.6   | 8.0   | 7.6   | 6.7   | 5.8   | 5.4   | (37) |
| (38) |                                                                                  | 5% WB       | MADB  | 13.1  | 11.7  | 11.9  | 11.7  | 11.5  | 9.3   | 10.0  | 12.8  | 13.2  | 13.2  | 13.2  | 12.5  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.5   | 4.3   | 5.2   | 5.1   | 5.9   | 5.8   | 6.3   | 7.7   | 7.1   | 6.3   | 5.3   | 5.0   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |       | 0.349 | 0.349 | 0.325 | 0.307 | 0.291 | 0.286 | 0.281 | 0.290 | 0.320 | 0.330 | 0.347 | 0.340 | (40) |
| (41) |                                                                                  | taud        |       | 2.507 | 2.529 | 2.586 | 2.595 | 2.599 | 2.615 | 2.614 | 2.561 | 2.465 | 2.474 | 2.473 | 2.518 | (41) |
| (42) |                                                                                  | Ebn at Noon |       | 989   | 970   | 961   | 928   | 895   | 875   | 899   | 932   | 949   | 978   | 985   | 1001  | (42) |
| (43) |                                                                                  | Edh at Noon |       | 114   | 107   | 95    | 84    | 75    | 69    | 72    | 85    | 105   | 112   | 117   | 113   | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |       | 7.70  | 6.83  | 5.79  | 4.61  | 3.48  | 2.73  | 3.09  | 4.04  | 5.42  | 6.53  | 7.16  | 7.79  | (44) |
| (45) |                                                                                  | RadStd      |       | 0.40  | 0.48  | 0.39  | 0.29  | 0.19  | 0.19  | 0.18  | 0.25  | 0.33  | 0.46  | 0.68  | 0.38  | (45) |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (9 neighbors) |  | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -79     | +179    | +123    |

Nomenclature: See separate page